

Bruce Chen

Senior Software Engineer · Boston, MA

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SUMMARY

Senior software engineer with 8+ years across full-stack development: UI, backend services, software design patterns, and database design. Currently focused on AI-assisted engineering workflows, designing where agents add real value, building reusable agent skills, and pushing the agent-as-orchestrator model into real dev work.

EXPERIENCE

MathWorks — Senior Software Engineer, Simulink Test Team Oct 2022 – Present · Natick, MA

- **[Current focus]** Delivering end-to-end V&V capabilities for the released **Simulink Agentic Toolkit** via skill-based and script-based agentic workflows with engineer-in-the-loop design and iterative feedback loops modeled on the human V&V cycle, focused on Simulink Test workflows like requirement-based test authoring and test interface setup
- Delivering test interface authoring for the next-generation **Simulink Test Manager V2** (parameter overrides, external inputs, source blocks, and signal visualization), driving test simulation runs for safety-critical model-based design
- Independently designed and delivered the **RoadRunner test case feature** in Simulink Test, integrating MathWorks' ADAS/AV scenario platform with the Simulink model-based V&V pipeline to enable scenario simulation and full V&V workflows for ADAS/AV customers

Hippo Insurance — Software Engineer Nov 2021 – Sep 2022 · Palo Alto, CA

- Refactored and extended underwriting backend logic powering the customer-facing quoting flow
- Extended internal tooling for state-specific home insurance regulation data

MathWorks — Software Engineer, MATLAB Test Team Mar 2020 – Oct 2021 · Natick, MA

- Built custom test plugins enabling real-time retrieval of execution details across serial and parallel test runs
- Built workflow tooling enabling power users to manage and scale large test suites

Eli Lilly (via Onebridge) — Software Engineer, Contract Jul 2018 – Feb 2020 · San Diego, CA

- Built backend data pipelines and APIs for an internal scientific data management platform, including PostgreSQL schema design and indexing for high-volume datasets
- Built web-based visualizations for molecule sequence analysis and alignment

SKILLS

Languages & Frameworks: C++, JavaScript (Node.js), Java, Python (Django)

Databases: PostgreSQL (relational DB design, schema optimization, indexing)

Model-Based Design: MATLAB, Simulink

AI & Agentic Tooling: Claude Code: skill design, meta-workflow scaffolding, agentic workflow architecture

EDUCATION

Brandeis University — M.S. Computer Science

Aug 2016 – May 2018 · Waltham, MA